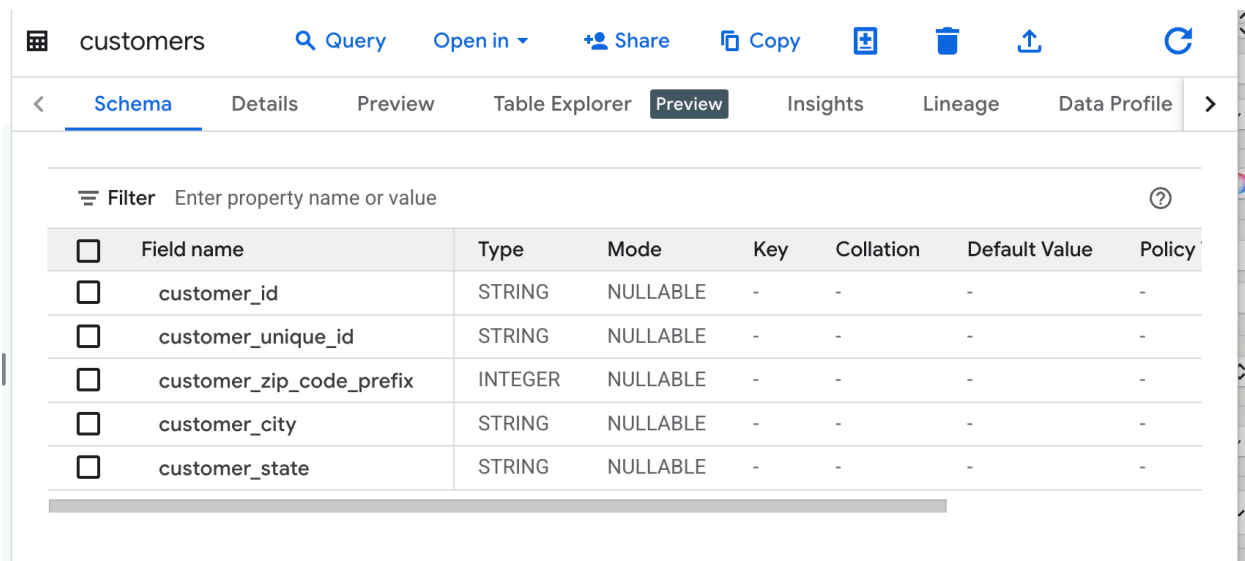


I. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

The Customers table consists of five columns. 'customer_id' column which is a string, 'customer_unique_id' which is also a string, 'customer_zip_code_prefix' it is a Integer value, 'customer_city' which represents the city name is a string and 'customer_state' which represents the state name is a string. These all five columns make 'customers' table.



The screenshot shows a database interface for a table named 'customers'. The interface includes a top navigation bar with options like 'Query', 'Open in', 'Share', 'Copy', and 'Preview'. Below this, there's a tabbed interface with 'Schema' selected. The 'Schema' tab displays a table with columns: Field name, Type, Mode, Key, Collation, Default Value, and Policy. The table lists five columns: customer_id (STRING, NULLABLE), customer_unique_id (STRING, NULLABLE), customer_zip_code_prefix (INTEGER, NULLABLE), customer_city (STRING, NULLABLE), and customer_state (STRING, NULLABLE).

Field name	Type	Mode	Key	Collation	Default Value	Policy
customer_id	STRING	NULLABLE	-	-	-	-
customer_unique_id	STRING	NULLABLE	-	-	-	-
customer_zip_code_prefix	INTEGER	NULLABLE	-	-	-	-
customer_city	STRING	NULLABLE	-	-	-	-
customer_state	STRING	NULLABLE	-	-	-	-

2. Get the time range between which the orders were placed.

A number of orders were placed over a period of two years, with the first order on September 4, 2016, at 21:15:19 UTC, and the last order on October 17, 2018, at 17:30:18 UTC.

target x *Untitled...ery x customers x orders x *Untitled...ery x

Untitled query Run Schedule Open in More Save Download Share

```

1
2 SELECT
3 | customer_id,order_purchase_timestamp, MIN(order_purchase_timestamp) OVER(ORDER BY order_purchase_timestamp ) first_order, MAX
4 | (order_purchase_timestamp) OVER(ORDER BY order_purchase_timestamp DESC) last_order
5 FROM `scaler-dsml-sql-456002.target.orders`
6
7

```

Query completed

Query results Save results Open in

Job information	Results	Chart	JSON	Execution details	Execution graph
Row	customer_id	order_purchase_timestamp	first_order	last_order	
1	a4b417188adddbc05b26b72d5...	2018-10-17 17:30:18 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
2	856336203359aa6a61bf3826f...	2018-10-16 20:16:02 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
3	4c2ec60c29d10c34bd49cb88a...	2018-10-03 18:55:29 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
4	bf6181a85bbb4115736c0a8db...	2018-10-01 15:30:09 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
5	2823ffda607a2316375088e0d...	2018-09-29 09:13:03 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
6	55c9dad94ec1a2ba57998bdb3...	2018-09-26 08:40:15 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
7	7a8a02e54cf3472df427e927f8...	2018-09-25 11:59:18 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
8	191984a8ba4cbb2145acb4fe3...	2018-09-20 13:54:16 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
9	ad2eb5d02c0808bcd8743117f...	2018-09-17 17:21:16 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
10	6ddc45ad08e6a2c885ca2255f...	2018-09-13 09:56:12 UTC	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	

Results per page: 50 1 - 50 of 99441

3. Count the Cities & States of customers who ordered during the given period.

There are a total of 4,119 cities in Brazil where Target has clients, and all of these 4119 cities are located within 27 states.

Untitled query Run Save Download Share

```

1 SELECT COUNT(DISTINCT c.customer_city) AS city_count, COUNT(DISTINCT c.customer_state) AS
2 state_count
3 FROM `target.customers` c
4 JOIN
5 (WITH order_range AS
6 | customer_id,order_purchase_timestamp, MIN(order_purchase_timestamp) OVER(ORDER BY
7 | order_purchase_timestamp ) first_order, MAX(order_purchase_timestamp) OVER(ORDER BY
8 | order_purchase_timestamp) last_order
9 FROM `scaler-dsml-sql-456002.target.orders`)
10 SELECT customer_id FROM order_range
11 WHERE order_purchase_timestamp BETWEEN first_order AND last_order) az
12 ON c.customer_id = az.customer_id
13

```

Query completed

✓ Query completed			
Query results		Save results ▾	Open in ▾
Job information	Results	Chart	JSON
Execution details			
Execution graph			
Row	city_count ▾	state_count ▾	
1	4119	27	

2. In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?

There is clear evidence of a strong upward trend in the number of orders placed over the past few years, reflecting growing customer engagement and business expansion. In 2016, from September 4th to the end of the year, only 329 orders were recorded. In 2017, total orders surged to 45,101. By just October 17th, 2018, the number of orders had already reached 54,011, surpassing the entire previous year’s total. This pattern demonstrates consistent year-over-year growth, suggesting effective operational strategies and increasing market demand.

Actionable Insights & Recommendations:

- The current growth trajectory reflects effective execution. Continue investing in successful initiatives such as marketing, logistics, and customer service — while making incremental optimizations based on data insights and market trends.
- Ensure the scalability of key systems (e.g., supply chain, warehousing, delivery network, customer support) to accommodate increased order volumes without compromising service quality.
- Leverage the growing customer base to build brand loyalty through referral programs, exclusive memberships, or early access to new products — helping to turn new buyers into repeat customers.
- While the trend is currently positive, regularly review metrics for signs of market saturation or slowing growth in mature segments to adjust strategies proactively.

customers

orders

*Untitled...ery

*Untitled...ery

Untitled query

Run

Save

Download

Share

```
1
2 SELECT EXTRACT(YEAR FROM order_purchase_timestamp) AS year, COUNT(order_id) AS cnt
3 FROM `scaler-dsml-sql-456002.target.orders`
4 GROUP BY EXTRACT(YEAR FROM order_purchase_timestamp)
5 ORDER BY year
6
7
8
9
10
```

Query completed

Query results

Save results

Open in

Job information

Results

Chart

JSON

Execution details

Execution graph

Row	year	cnt
1	2016	329
2	2017	45101
3	2018	54011

Results per page:

50

1 – 3 of 3

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Yes, the data indicates clear signs of monthly seasonality in the number of orders placed, particularly when analyzing year-over-year patterns from 2016 to 2018. In 2017, order volumes were lower in the first quarter (January–April), followed by a steady increase from mid-year, culminating in a peak during November with 7,544 orders — likely influenced by year-end shopping events and holidays. In 2018, monthly order volumes remained strong from January to August, suggesting sustained consumer activity. However, there was a sharp decline starting in September, and orders almost completely dropped off by October, which may reflect seasonal transitions or operational changes.

Actionable Insights & Recommendations:

- Plan major marketing campaigns, product launches, and promotions to coincide with months showing strong order volumes — especially November and mid-year months (June–August). Increase inventory and staffing during these peak periods to ensure timely fulfillment and positive customer experiences.

- Introduce incentives such as loyalty rewards, clearance sales, or limited-time discounts in traditionally low-activity months like January–March and September–October. Promote bundled offers or themed campaigns like "New Year Essentials" or "Back-to-Work Deals" to spark customer interest and counteract seasonal lulls.
- Conduct further analysis to understand the sharp decline in orders after August 2018. Potential factors to investigate include Changes in website/app usability, Logistics or delivery issues, Shifts in customer behaviour or marketing performance.
- Launch early-access promotions or loyalty-based perks in September and October to ramp up engagement leading into the busy holiday season.

et X *Untitled...ery X customers X orders X *Untitled...ery X *Untitled...ery X

Untitled query Run Schedule Open in More Save Download

```
1
2 SELECT EXTRACT(MONTH FROM order_purchase_timestamp) AS month_,
3       SUM(CASE
4         | WHEN EXTRACT (YEAR FROM order_purchase_timestamp) = 2016 THEN 1 ELSE 0
5       END) AS year_2016,
6       SUM(CASE
7         | WHEN EXTRACT(YEAR FROM order_purchase_timestamp) = 2017 THEN 1 ELSE 0
8       END) AS year_2017,
9       SUM(CASE
10        | WHEN EXTRACT(YEAR FROM order_purchase_timestamp) = 2018 THEN 1 ELSE 0
11        END) AS year_2018
12 FROM `scaler-dsml-sql-456002.target.orders`
13
14 GROUP BY EXTRACT(MONTH FROM order_purchase_timestamp)
15 ORDER BY month_
16
```

This query will process 776.88 KB when run

Query results [Save results](#)

Job information	Results	Chart	JSON	Execution details	Execution graph
Row	month_	year_2016	year_2017	year_2018	
1	1	0	800	7269	
2	2	0	1780	6728	
3	3	0	2682	7211	
4	4	0	2404	6939	
5	5	0	3700	6873	
6	6	0	3245	6167	
7	7	0	4026	6292	
8	8	0	4331	6512	
9	9	4	4285	16	
10	10	324	4631	4	
11	11	0	7544	0	
12	12	1	5673	0	

3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- **0-6 hrs : Dawn**
- **7-12 hrs : Mornings**
- **13-18 hrs : Afternoon**
- **19-23 hrs : Night**

An evaluation of order timestamps reveals that Afternoon (13:00–18:00) is the most popular time for placing orders, with a total of 38,135 orders. This aligns with the typical daily routine when customers are more active and likely to browse or shop online. The Night period (19:00–23:00) follows closely with 28,331 orders, suggesting that many customers engage in online shopping after work hours or before going to bed. The Morning period (07:00–12:00) also sees strong activity, with 27,733 orders. Unsurprisingly, Dawn (00:00–06:00) has the lowest engagement, with only 5,242 orders, likely due to people being asleep or inactive during these hours.

Actionable Insights & Recommendations:

- Schedule major marketing campaigns, product launches, and time-sensitive discounts to align with peak order times - particularly during Afternoon and Night hours, when customer activity is highest. Send push notifications, email reminders, and personalized offers around these periods to maximize engagement and conversion.
- To increase order volume during the Dawn period, consider launching early-hour exclusive deals or flash sales starting at midnight or 1:00 AM. We can target Night owls, Shift workers, Holiday shoppers looking for early access.
- Utilize the low-traffic Dawn window for clearance events or limited-stock promotions, helping move older or seasonal inventory without competing with high-demand times.
- Optimize digital ad spend by increasing visibility during high-traffic time blocks and experimenting with low-cost bidding during Dawn, targeting niche segments that might convert in those hours.

Untitled queryRunScheduleOpen inMoreSaveDownloadShare

```
1
2 WITH ordered_timings AS
3 (SELECT customer_id,
4  CASE
5  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
6  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Mornings'
7  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'
8  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 19 AND 23 THEN 'Night'
9  END AS day_div
10 FROM `target.orders`)
11
12 SELECT day_div, COUNT(customer_id) AS no_of_orders
13 FROM ordered_timings
14 GROUP BY day_div
15 ORDER BY day_div
16
```

This query will process 3.98 MB when run

Query results				Save results	
Job information		Results	Chart	JSON	Execution details
Execution graph					
Row	day_div	no_of_orders			
1	Afternoon	38135			
2	Dawn	5242			
3	Mornings	27733			
4	Night	28331			

Results per page: 501 – 4 of 4

3. Evolution of E-commerce orders in the Brazil region:

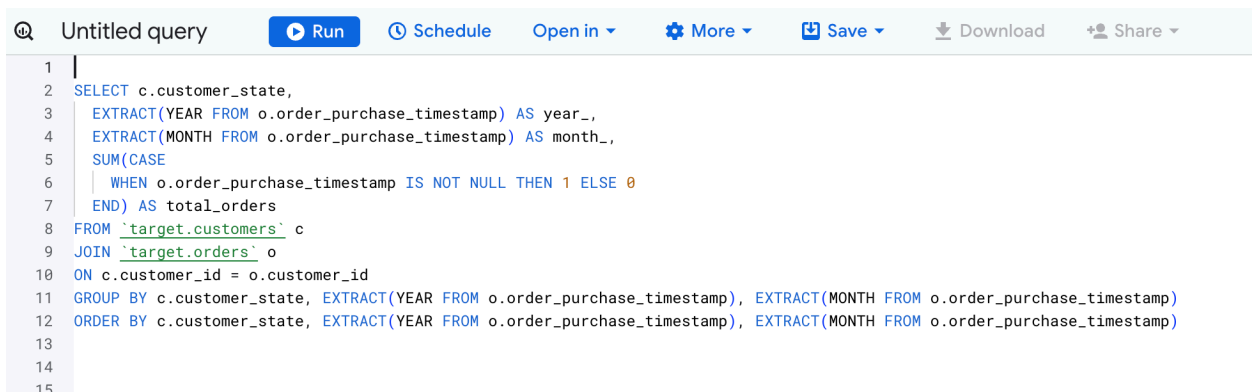
1. Get the month on month no. of orders placed in each state.

A month-on-month breakdown of order volumes by state reveals significant disparities. Certain states consistently report around 3,200 orders per month, demonstrating seasonal stability and strong customer engagement. Conversely, some states report very low order volumes—as few as 2 or 4 orders per month, and in some cases, no orders at all during certain months.

This imbalance suggests a need for further investigation into the factors influencing regional engagement, including customer awareness, product relevance, and logistical accessibility.

Actionable Insights & Recommendations:

- Conduct market research in states with low or no order volume to understand customer behaviour, online shopping adoption barriers, and local preferences. Consider surveys or focus groups to gain qualitative insights.
- In regions with limited online activity, run targeted awareness campaigns that emphasize the benefits of online shopping with Target, such as, Exclusive online deals and discounts, broader range of products compared to local stores, Convenience of home delivery and returns.
- Analyze product preferences across different regions. If underperforming states show lower interest due to product irrelevance, collaborate with local vendors or brands to add inventory that aligns with regional tastes, needs, and cultural relevance.
- Offer first-time shopper incentives in low-performing states—such as free delivery, welcome discounts, or referral rewards—to encourage initial trial and reduce resistance to online purchases.



```
1 |
2 | SELECT c.customer_state,
3 |     EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year_,
4 |     EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month_,
5 |     SUM(CASE
6 |         WHEN o.order_purchase_timestamp IS NOT NULL THEN 1 ELSE 0
7 |     END) AS total_orders
8 | FROM `target.customers` c
9 | JOIN `target.orders` o
10 | ON c.customer_id = o.customer_id
11 | GROUP BY c.customer_state, EXTRACT(YEAR FROM o.order_purchase_timestamp), EXTRACT(MONTH FROM o.order_purchase_timestamp)
12 | ORDER BY c.customer_state, EXTRACT(YEAR FROM o.order_purchase_timestamp), EXTRACT(MONTH FROM o.order_purchase_timestamp)
13 |
14 |
15 |
```


Query results						Save results
Job information		Results	Chart	JSON	Execution details	Execution graph
Row	customer_state ▼	year_ ▼	month_ ▼	total_orders ▼		
1	AC	2017	1	2		
2	AC	2017	2	3		
3	AC	2017	3	2		
4	AC	2017	4	5		
5	AC	2017	5	8		
6	AC	2017	6	4		
7	AC	2017	7	5		
8	AC	2017	8	4		
9	AC	2017	9	5		
10	AC	2017	10	6		
11	AC	2017	11	5		
12	AC	2017	12	5		
13	AC	2018	1	6		
14	AC	2018	2	3		
15	AC	2018	3	2		

2. How are the customers distributed across all the states?

Brazil is composed of 27 states, and customers are distributed across all of them. However, there is a significant variation in customer distribution. The state of Roraima (RR) has the fewest customers, with only 46 registered users, whereas São Paulo (SP) leads by a wide margin with 41,746 customers. This indicates a strong regional concentration of customer activity, with some states vastly outperforming others in terms of engagement.

Actionable Insights & Recommendations:

- The sharp contrast between high- and low-performing states reveals uneven market penetration. States with lower customer engagement, like Roraima (RR), should be the focus of localized marketing efforts, including promotional campaigns tailored to regional preferences and cultural nuances.
- Low customer density in certain states may correlate with limited delivery infrastructure or slower fulfillment capabilities. Conduct a logistics audit in these areas and explore partnerships with local delivery services to improve accessibility and delivery times.

- Analyze top-performing states like São Paulo (SP) to understand what drives engagement, such as popular product categories, shipping efficiency, or promotional effectiveness. Use these insights to replicate successful strategies in underperforming regions.
- Implement geo-targeted advertising, first-time shopper discounts, or free delivery promotions specifically for low-engagement states. This will help increase visibility and drive initial trials in those markets.

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customers ▾ ×
orders ▾ ×
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Untitled query
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Schedule
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```

1
2 SELECT customer_state, COUNT(customer_id) AS state_cust_count
3 FROM `target.customers`
4 GROUP BY customer_state
5 ORDER BY customer_state
6

```

✓ This query will process 3.6 MB when run.

Query results

Job information		Results	Chart	JSON	Execution details	Execution graph
Row	customer_state ▾	state_cust_count ▾				
1	AC	81				
2	AL	413				
3	AM	148				
4	AP	68				
5	BA	3380				
6	CE	1336				
7	DF	2140				
8	ES	2033				
9	GO	2020				
10	MA	747				
11	MG	11635				

Results per

4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only). You can use the "payment_value" column in the payments table to get the cost of orders.

The total payment received between January and August of 2017 was 3.63 million, while for the same period in 2018, it increased to 8.62 million. This represents a 137.73% growth in the cost of orders from 2017 to 2018.

This significant increase indicates that the business experienced exponential growth over the two-year period, suggesting that the current business practices are effective and should be continued.

```
1 |
2 | WITH month_sorting AS
3 | (SELECT SUM(p.payment_value) AS payment_value,
4 |  EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year_
5 | FROM `target.payments` p
6 | JOIN `target.orders` o
7 | ON p.order_id = o.order_id
8 | WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017,2018) AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
9 |   AND o.order_status <> 'canceled'
10 | GROUP BY EXTRACT(YEAR FROM o.order_purchase_timestamp)
11 | )
12 | SELECT
13 |   MAX(CASE WHEN year_ = 2017 THEN payment_value END) AS payment_year_2017,
14 |   MAX(CASE WHEN year_ = 2018 THEN payment_value END) AS payment_year_2018,
15 |   ROUND((MAX(CASE WHEN year_ = 2018 THEN payment_value END) - MAX(CASE WHEN year_ = 2017 THEN payment_value END)) / MAX(CASE WHEN year_
16 |   = 2017 THEN payment_value END) * 100, 2) AS percent_increase
17 | FROM month_sorting
```

✓ This query will process 9.19 MB when run.

Query results

Save results Open in

Job information	Results	Chart	JSON	Execution details	Execution graph
Row	payment_year_2017	payment_year_2018	percent_increase		
1	3627841.559999...	8624464.940000...	137.73		

2. Calculate the Total & Average value of order price for each state.

Out of the 27 states in Brazil, "SP" is the state with the highest total order amount, approximately 5.2 million. The state with the lowest total order value is "RR", with just 7829.43. While "SP" leads in total sales. in contrast it is the state with lowest average order value with just 109.65. Meanwhile The state with highest average order value is "PB" with 191.48.

```

2 SELECT
3   c.customer_state,
4   ROUND(SUM(oi.price), 2) AS total_order_value, ROUND(AVG(oi.price), 2) AS avg_order_value
5 FROM `target.orders` o
6 JOIN `target.order_items` oi ON o.order_id = oi.order_id
7 JOIN `target.customers` c ON o.customer_id = c.customer_id
8 GROUP BY c.customer_state
9 ORDER BY avg_order_value DESC

```

✓ This query will process 14.56 MB when run.

Query results

[Save](#)

Job information		Results	Chart	JSON	Execution details	Execution graph
Row	customer_state ▼	total_order_value ▼	avg_order_value ▼			
1	SP	5202955.05	109.65			
2	RJ	1824092.67	125.12			
3	MG	1585308.03	120.75			
4	RS	750304.02	120.34			
5	PR	683083.76	119.0			
6	SC	520553.34	124.65			
7	BA	511349.99	134.6			
8	DF	302603.94	125.77			
9	GO	294591.95	126.27			
10	ES	275037.31	121.91			

3. Calculate the Total & Average value of order freight for each state.

Out of the 27 states in Brazil, "SP" is the state with the highest total freight amount with 718723.07. The state with the lowest total freight amount is "RR", with just 2235.19. While "SP" leads in total freight value, in contrast it is the state with lowest average freight value of 15.15. The state with highest average freight value is "RR" with 42.98.

```
2 SELECT
3 | c.customer_state, ROUND(SUM(oi.freight_value), 2) AS total_freight_value, ROUND(AVG(oi.freight_value), 2) AS avg_freight_value
4 FROM `target.orders` o
5 JOIN `target.order_items` oi ON o.order_id = oi.order_id
6 JOIN `target.customers` c ON o.customer_id = c.customer_id
7 GROUP BY c.customer_state
8 ORDER BY total_freight_value DESC
```

✔ This query will process 14.56 MB when run.

Query results

Save results

Open in

Job information		Results	Chart	JSON	Execution details	Execution graph
Row	customer_state	total_freight_value	avg_freight_value			
1	SP	718723.07	15.15			
2	RJ	305589.31	20.96			
3	MG	270853.46	20.63			
4	RS	135522.74	21.74			
5	PR	117851.68	20.53			
6	BA	100156.68	26.36			
7	SC	89660.26	21.47			
8	PE	59449.66	32.92			
9	GO	53114.98	22.77			
10	DF	50625.5	21.04			

5. Analysis based on sales, freight and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

$$\text{time_to_deliver} = \text{order_delivered_customer_date} - \text{order_purchase_timestampdiff_}$$

$$\text{estimated_delivery} = \text{order_delivered_customer_date} - \text{order_estimated_delivery_date}$$

Out of the total orders, 96,476 were successfully delivered to customers, while the remaining 2,965 orders are still in transit. The delivery time varied by order ID, influenced by factors such as distance, transportation frequency, climate conditions, and other logistical variables. Among the delivered orders, only 6,535 (approximately 6.8%) exceeded the estimated delivery time, while the majority were delivered on or before the estimated date, indicating generally strong performance in logistics and fulfillment.

Actionable Insights & Recommendations:

- With only a small portion of orders exceeding estimated delivery dates, delivery operations are mostly efficient. However, the 6,535 delayed orders should be

analyzed in detail to identify common factors, such as delivery routes, warehouses, or specific product types, that may be contributing to delays.

- If delays are concentrated in specific regions or tied to certain product categories, implement targeted logistics optimizations like alternate carriers, localized warehousing, or adjusted inventory management to reduce delays.
- In regions where orders are consistently delivered on or before the estimated date, consider slightly reducing the estimated delivery window. This can enhance customer satisfaction by aligning perceived and actual performance and encourage repeat purchases.
- If delivery tracking is not in use then consider adding or enhancing real-time delivery tracking and notifications, which can reduce anxiety and inquiries related to in-transit orders.

Untitled query

RunScheduleOpen inMoreSaveDownloadShare

1

2

3SELECT order_id,

4order_purchase_timestamp,

5order_delivered_customer_date,

6order_estimated_delivery_date,

7DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS time_to_deliver,

8DATE_DIFF(order_delivered_customer_date, order_estimated_delivery_date, DAY) AS diff_estimated_delivery

9FROM `target.orders`

10WHERE order_delivered_customer_date IS NOT NULL

11ORDER BY order_id

12

13

14

✓

This query will process 5.48 MB when run.

Row	order_id	order_purchase_timestamp	order_delivered_customer_date	order_estimated_delivery_date	time_to_deliver	diff_estimated_delivery
1	00010242fe8c5a6d1ba2dd792...	2017-09-13 08:59:02 UTC	2017-09-20 23:43:48 UTC	2017-09-29 00:00:00 UTC	7	-8
2	00018f77f2f0320c557190d7a...	2017-04-26 10:53:06 UTC	2017-05-12 16:04:24 UTC	2017-05-15 00:00:00 UTC	16	-2
3	000229ec398224ef6ca0657da...	2018-01-14 14:33:31 UTC	2018-01-22 13:19:16 UTC	2018-02-05 00:00:00 UTC	7	-13
4	00024acbcdf0a6daa1e931b03...	2018-08-08 10:00:35 UTC	2018-08-14 13:32:39 UTC	2018-08-20 00:00:00 UTC	6	-5
5	00042b26cf59d7ce69dfabb4e...	2017-02-04 13:57:51 UTC	2017-03-01 16:42:31 UTC	2017-03-17 00:00:00 UTC	25	-15
6	00048cc3ae777c65dbb7d2a06...	2017-05-15 21:42:34 UTC	2017-05-22 13:44:35 UTC	2017-06-06 00:00:00 UTC	6	-14
7	00054e8431b9d7675808bcb8...	2017-12-10 11:53:48 UTC	2017-12-18 22:03:38 UTC	2018-01-04 00:00:00 UTC	8	-16
8	000576fe39319847cbb9d288c...	2018-07-04 12:08:27 UTC	2018-07-09 14:04:07 UTC	2018-07-25 00:00:00 UTC	5	-15
9	0005a1a1728c9d785b8e2b08...	2018-03-19 18:40:33 UTC	2018-03-29 18:17:31 UTC	2018-03-29 00:00:00 UTC	9	0
10	0005f50442cb953dcd1d21e1f...	2018-07-02 13:59:39 UTC	2018-07-04 17:28:31 UTC	2018-07-23 00:00:00 UTC	2	-18
11	00061f2a7bc09da83e415a52d...	2018-03-24 22:16:10 UTC	2018-03-29 00:04:19 UTC	2018-04-09 00:00:00 UTC	4	-10
12	00063b381e2406b52ad42947...	2018-07-27 17:21:27 UTC	2018-08-07 13:56:52 UTC	2018-08-07 00:00:00 UTC	10	0
13	0006ec9db01a64e59a68b2c34...	2018-07-24 17:04:17 UTC	2018-07-31 01:04:15 UTC	2018-08-22 00:00:00 UTC	6	-21
14	0008288aa423d2a3f0fcb17c...	2018-02-13 22:10:21 UTC	2018-02-26 13:55:22 UTC	2018-03-06 00:00:00 UTC	12	-7
15	0009792311464db532ff765bf...	2018-08-14 20:43:09 UTC	2018-08-22 12:02:27 UTC	2018-08-28 00:00:00 UTC	7	-5

2. Find out the top 5 states with the highest & lowest average freight value.

Top 5 states with highest average freight values are 1. RO with 50.91, 2. CE with 46.3, 3. PB with 39.19, 4. PI with 36.94 and 5. AC with 32.84. The 5 states with lowest average freight values are 1. SP with 18.45, 2. PA with 19.39, 3. RJ with 19.4, 4. DF with 20.57, 5. PR with 22.72.

```
2 WITH state_freight AS
3 (SELECT s.seller_state AS state, ROUND(AVG(oi.freight_value), 2) AS avg_freight_value
4 FROM `target.order_items` oi
5 JOIN `target.sellers` s
6 ON oi.seller_id = s.seller_id
7 GROUP BY s.seller_state
8 ORDER BY avg_freight_value DESC
9 ),
10
11 rank_states AS
12 (SELECT state, avg_freight_value,
13     RANK() OVER(ORDER BY avg_freight_value DESC) AS top_5_freight_states,
14     RANK() OVER(ORDER BY avg_freight_value) AS least_5_freight_states
15 FROM state_freight)
```

```
16
17 SELECT state, avg_freight_value, top_5_freight_states
18 FROM rank_states
19 WHERE top_5_freight_states <= 5
20 ORDER BY top_5_freight_states
21
22
23
```

✓ Query completed

Query results

Job information		Results	Chart	JSON	Execution details	Execution graph
Row	state ▼	avg_freight_value ▼	top_5_freight_states			
1	RO	50.91	1			
2	CE	46.38	2			
3	PB	39.19	3			
4	PI	36.94	4			
5	AC	32.84	5			

```

1
2 WITH state_freight AS
3 (SELECT s.seller_state AS state, ROUND(AVG(oi.freight_value), 2) AS avg_freight_value
4 FROM `target.order_items` oi
5 JOIN `target.sellers` s
6 ON oi.seller_id = s.seller_id
7 GROUP BY s.seller_state
8 ORDER BY avg_freight_value DESC
9 ),
10
11 rank_states AS
12 (SELECT state, avg_freight_value,
13   RANK() OVER(ORDER BY avg_freight_value DESC) AS top_5_freight_states,
14   RANK() OVER(ORDER BY avg_freight_value) AS least_5_freight_states
15 FROM state_freight)
16
17 SELECT state, avg_freight_value, least_5_freight_states
18 FROM rank_states
19 WHERE least_5_freight_states <= 5
20 ORDER BY least_5_freight_states
21
22
23

```

Query results

Job information

Results

Chart

JSON

Execution details

Exec

Row	state	avg_freight_value	least_5_freight_state
1	SP	18.45	1
2	PA	19.39	2
3	RJ	19.47	3
4	DF	20.57	4
5	PR	22.72	5

3. Find out the top 5 states with the highest & lowest average delivery time.

Top 5 states with highest average delivery time are 1. RR with 4.02, 2. SE with 3.14, 3. MA with 3.12, 4. RN with 3.11 and 5. AP with 3.05. The 5 states with lowest average freight values are 1. AM with 2.31, 2. RO with 2.32, 3. SP with 2.6774, 4. MS with 2.6775, 5. GO with 2.679.


```

3 WITH delivery_time AS
4 (SELECT c.customer_state AS state,
5  AVG(DATE_DIFF(order_delivered_carrier_date, order_purchase_timestamp, DAY)) AS avg_delivery_time
6  FROM `target.orders` o
7  JOIN `target.customers` c
8  ON o.customer_id = c.customer_id
9  WHERE order_delivered_carrier_date IS NOT NULL
10 GROUP BY c.customer_state
11 ),
12
13 states_ranked AS
14 (SELECT state, avg_delivery_time,
15  RANK() OVER(ORDER BY avg_delivery_time) AS states_least_avg_delivery_time,
16  RANK() OVER(ORDER BY avg_delivery_time DESC) AS states_highest_avg_delivery_time
17 FROM delivery_time)
18
19 SELECT state, avg_delivery_time, states_highest_avg_delivery_time
20 FROM states_ranked
21 WHERE states_highest_avg_delivery_time <= 5
22 ORDER BY states_highest_avg_delivery_time
23
24

```

Query results

Job information		Results	Chart	JSON	Execution details	Execution g
Row	state ▼	avg_delivery_time ▼	states_highest_avg_c			
1	RR	4.022222222222...	1			
2	SE	3.148255813953...	2			
3	MA	3.127891156462...	3			
4	RN	3.110187110187...	4			
5	AP	3.044776119402...	5			

```

2
3 WITH delivery_time AS
4 (SELECT c.customer_state AS state,
5    AVG(DATE_DIFF(order_delivered_carrier_date, order_purchase_timestamp, DAY)) AS avg_delivery_time
6 FROM `target.orders` o
7 JOIN `target.customers` c
8 ON o.customer_id = c.customer_id
9 WHERE order_delivered_carrier_date IS NOT NULL
10 GROUP BY c.customer_state
11 ),
12
13 states_ranked AS
14 (SELECT state, avg_delivery_time,
15    RANK() OVER(ORDER BY avg_delivery_time) AS states_least_avg_delivery_time,
16    RANK() OVER(ORDER BY avg_delivery_time DESC) AS states_highest_avg_delivery_time
17 FROM delivery_time)
18
19 SELECT state, avg_delivery_time, states_least_avg_delivery_time
20 FROM states_ranked
21 WHERE states_least_avg_delivery_time <= 5
22 ORDER BY states_least_avg_delivery_time
23
24

```

Query results

Job information		Results	Chart	JSON	Execution details	Execution gra
Row	state ▼	avg_delivery_time ▼	states_least_avg_del			
1	AM	2.312925170068...	1			
2	RO	2.329218106995...	2			
3	SP	2.677443829849...	3			
4	MS	2.677556818181...	4			
5	GO	2.679235796882...	5			

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery. You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Out of 27 states in Brazil, the top 5 states with highest order delivery times are

1. AC with 19.76,
2. RO with 19.13,
3. AP with 18.73,
4. AM with 18.61 and
5. RR with 16.41.

Untitled query

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```
1
2 SELECT c.customer_state AS state,
3        ROUND(AVG(DATE_DIFF(o.order_estimated_delivery_date, o.order_delivered_customer_date, DAY)), 2) as early_by_days
4 FROM `target.orders` o
5 JOIN `target.customers` c
6 ON o.customer_id = c.customer_id
7 WHERE o.order_delivered_customer_date IS NOT NULL
8 GROUP BY c.customer_state
9 ORDER BY early_by_days DESC
10 LIMIT 5
11
12
```

Query completed

Query results

Save resultsOptions

Job informationResultsChartJSONExecution detailsExecution graph

Row	state	early_by_days
1	AC	19.76
2	RO	19.13
3	AP	18.73
4	AM	18.61
5	RR	16.41

6. Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

The analysis of the report reveals that credit cards are the dominant payment type compared to other methods. Credit cards consistently remained the most preferred mode of payment across all observed months. For instance, in September 2017, there were 3,283 credit card transactions, significantly higher than other payment types such

In contrast, voucher and debit card usage declined over time. Voucher transactions dropped from 287 orders in September 2017 to just 4 orders by October 2018. Debit card usage remained consistently low throughout the observed period. The "not_defined" payment type appeared rarely, with only 2 orders in August 2018 and 1 in September 2018.

- Since credit cards dominate across all months, we can partner with banks to offer more credit card promotions, such as cash-back offers, no-cost EMI plans, and exclusive discounts.
- With UPI usage steadily increasing, it's recommended to offer UPI-specific discounts or rewards to encourage further adoption and enhance the mobile checkout experience.
- Assess whether voucher-based campaigns are generating a meaningful return on investment (ROI). If not, consider reallocating the budget to more effective incentives or channels.

Untitled query

Run

Schedule

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```

2 SELECT FORMAT_DATETIME('%Y-%m', order_purchase_timestamp) AS month_year, p.payment_type, COUNT(o.order_id) AS no_of_orders
3 FROM `target.payments` p
4 JOIN `target.orders` o
5 ON p.order_id = o.order_id
6 GROUP BY FORMAT_DATETIME('%Y-%m', order_purchase_timestamp), p.payment_type
7 ORDER BY month_year, p.payment_type

```

Query completed

Query results

Save results

Open in

Job information

Results

Chart

JSON

Execution details

Execution graph

Row	month_year	payment_type	no_of_orders
1	2016-09	credit_card	3
2	2016-10	UPI	63
3	2016-10	credit_card	254
4	2016-10	debit_card	2
5	2016-10	voucher	23
6	2016-12	credit_card	1
7	2017-01	UPI	197
8	2017-01	credit_card	583
9	2017-01	debit_card	9
10	2017-01	voucher	61
11	2017-02	UPI	398

Results per page:

50

1 – 50 of 90

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2. Find the no. of orders placed on the basis of the payment installments that have been paid.

The analysis of the payment installments field reveals that customers have utilized a wide range of installment options, from 1 to 24 payments. The data indicates: The most common installment option is 12 installments, accounting for 133 orders. Other popular choices include 15 installments (74 orders) and 18 installments (27 orders). A gradual decline is observed in the number of orders as the number of installments increases beyond 12. Installment options such as 21, 22, and 23 are used very rarely, with only 1 to 3 orders each. The presence of orders using installment plans up to 24 months highlights that long-term financing is available, though rarely used.

Actionable Insights & Recommendations:

- Since 12-month installments are the most popular, emphasize this option in product pages and promotions to improve conversions.
- Installment options beyond 18 months (e.g., 21–24 months) have extremely low adoption. Consider removing them if they add unnecessary complexity or cost.
- If the business aims to promote shorter installment periods to improve cash flow then offer discounts for 6-month or 9-month plans, Interest-free payment plans for shorter terms.

Untitled query

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```
2 SELECT payment_installments, COUNT(*) AS num_orders
3 FROM `target.payments`
4 WHERE payment_installments > 0
5 GROUP BY payment_installments
6 ORDER BY payment_installments
```

This query will process 811.61 KB when run.

Query results

Save results

Job informationResultsChartJSONExecution detailsExecution graph

Row	payment_installment	num_orders
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	5	5239
6	6	3920
7	7	1626
8	8	4268
9	9	644
10	10	5328
11	11	23
12	12	133

Results per page501 - 23 of 23

Job information			Results	Chart	JSON	Execution details	Execution graph
Row	payment_installment	num_orders					
12	12	133					
13	13	16					
14	14	15					
15	15	74					
16	16	5					
17	17	8					
18	18	27					
19	20	17					
20	21	3					
21	22	1					
22	23	1					
23	24	18					

Results per na